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Exp.No-12

Title- STACK USING ARRAYS-PUSH(), POP()
AND DISPLAY() METHODS

```
In [23]: def push(stack, itemm, max_size):
    if len(stack)>= max_size:
        print("Stack Overflow")
    else:
        stack.append(item)
        print("Pushed", item, "to stack")
        return item
def pop(stack):
    if not stack:
        print("Stack underflow")
    else:
        item = stack.pop()
        print("Poped", item, "From stack")
        return item

def display(stack):
    if not stack:
        print("Stack is empty")
    else:
        print("Stack element are")
        for item in reversed(stack):
            print(item)

#input
max_size = int(input("Enter the maximum size of the stack: "))
stack = []

while True:
    print("=====")
    print("\n1. Push\n2. Pop\n3. Display\n4. Exit")
    choice = int(input("\n Enter your choice "))

    if choice == 1:
        item = int(input("Enter the item to push"))
        push(stack, item, max_size)
```

```
elif choice == 2:
    pop(stack)
elif choice == 3:
    display(stack)
elif choice == 4:
    break
else:
    print("Invalid choice, please try again...")
```

=====

1. Push
2. Pop
3. Display
4. Exit
Pushed 10 to stack

=====

1. Push
2. Pop
3. Display
4. Exit
Pushed 20 to stack

=====

1. Push
2. Pop
3. Display
4. Exit
Pushed 30 to stack

=====

1. Push
2. Pop
3. Display
4. Exit
Pushed 40 to stack

=====

1. Push
2. Pop
3. Display
4. Exit
Pushed 50 to stack

=====

1. Push
2. Pop
3. Display
4. Exit

Stack Overflow

=====

1. Push
2. Pop
3. Display
4. Exit

Stack element are

50
40
30
20
10

=====

1. Push
2. Pop
3. Display
4. Exit

Poped 50 From stack

=====

1. Push
2. Pop
3. Display
4. Exit

Poped 40 From stack

=====

1. Push
2. Pop
3. Display
4. Exit

Poped 30 From stack

=====

1. Push
2. Pop
3. Display
4. Exit

Poped 20 From stack

=====

1. Push
2. Pop
3. Display
4. Exit

Poped 10 From stack

=====

1. Push
2. Pop
3. Display
4. Exit

Stack underflow

=====

1. Push
2. Pop
3. Display
4. Exit

Stack is empty

=====

1. Push
2. Pop
3. Display
4. Exit